ADVANCED MACHINE LEARNING · WORKSHEET 17

Decision Trees Part Two

ID3 and controlling tree size

STUDENT COPY

Name:	**Roll No.:**	**Date:**

PART I From CART to ID3

In the previous Lecture 16, we studied how to apply the CART algorithm to a dataset. In this lecture, we will study a different algorithm, ID3, and understand another way of splitting a decision tree. We will then learn how pruning controls the size of a tree.

1 Recall what CART did

▷ HOOK

In the employment example, CART compared questions such as Degree = UG and Field = Science. It divided the rows into two groups, calculated a score for each question and chose the best split.

Step	**CART classification**	**CART regression**
Target	A class, such as Job Outcome	A number, such as Test Score
Score used in Lecture 16	Weighted child Gini	Weighted child MSE
Leaf prediction	The most common class	The mean of the scores
Shared process	Compare questions, divide the rows and repeat inside each child.	

What will change with ID3

We will again divide rows using their features. This time we will measure class mixture using **entropy** and choose the feature with the largest **information gain**. A categorical feature can create one branch for each of its values. For example, Degree can create three ID3 branches: UG, PG and PhD. The CART question Degree = UG created only two branches: UG and not UG. **Classroom check:** a leaf contains E, U, U. E means Employed and U means Unemployed. What should a majority-class leaf predict, and why?

♦ TAKEAWAY

Today: define ID3 → understand entropy → calculate information gain → build a classification tree → compare CART and ID3 → learn pre-pruning and post-pruning.

PART II ID3 using the employment data

Use the same ten employment records as Worksheet 16. Keep Degree and Field as features and Job Outcome as the target.

2 What is ID3

ID3 means Iterative Dichotomiser 3. It is an algorithm that builds a classification tree by choosing the feature with the largest information gain at each node. **Basic ID3 is used only for classification and works with categorical features. It does not directly split numerical features.** Average Grade is numerical, so we leave it out of this ID3 example.

Row	Degree	Field	Job Outcome
0	UG	Science	E
1	UG	Arts	U
2	PG	Science	E
3	PhD	Science	E
4	PG	Arts	U
5	PhD	Arts	E
6	UG	Science	U
7	PG	Arts	E
8	UG	Arts	U
9	PhD	Science	E

E = Employed; U = Unemployed. There are 6 E and 4 U in the complete dataset. The row number is a label for a record, not a predictor.

$p_E = \frac{6}{10}$, $p_U = \frac{4}{10}$

The process is similar but the score changes

CART in Lecture 16 chose the smallest weighted child Gini. ID3 chooses the largest information gain. To calculate information gain, we first need entropy.

Entropy measures how mixed the classes are

If every row has the same class, there is no uncertainty about a training row's class. If both classes are equally common, there is more uncertainty.

$$H(D) = -\sum_{k=1}^{K} p_k \log_2 p_k.$$

p_k is the fraction of rows in class k. For E and U, this becomes

$$H(D) = -p_E \log_2 p_E - p_U \log_2 p_U.$$

Use base-2 logarithms, so entropy is measured in bits. A class with probability 0 contributes 0. For two classes, a pure node has entropy 0 and an equally mixed node has entropy 1.

Worked example A small entropy dry run

Use rows 0, 1 and 8, whose labels are E, U and U.
1 Count: one E and two U.
2 Find proportions: $p_E = 1/3$ and $p_U = 2/3$.
3 Substitute: $H = -(1/3)\log_2(1/3) - (2/3)\log_2(2/3) = 0.9183$ bits.
Next, use these same rows to calculate information gain step by step.

Information Gain = Change in Entropy from Parent to child.

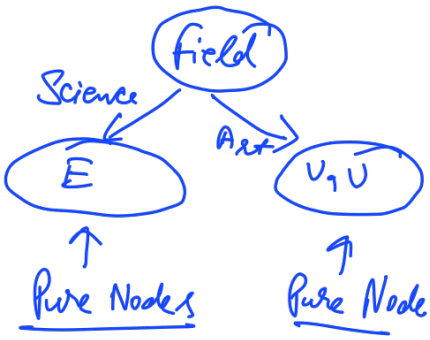
3 A small dry run from entropy to information gain

Start with just rows 0, 1 and 8 from the employment data. This small example helps us understand the calculation before comparing the complete ten-row dataset.

Row	Degree	Field	Outcome
0	UG	Science	E
1	UG	Arts	U
8	UG	Arts	U

Step 1 Count the classes

There is 1 E and there are 2 U. Therefore $p_E = 1/3$ and $p_U = 2/3$.



Step 2 Calculate the parent entropy

$$H(D) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3}.$$

Using $\log_2(1/3) = -1.5850$ and $\log_2(2/3) = -0.5850$,

$$H(D) = 0.5283 + 0.3900 = 0.9183 \text{ bits.}$$

Step 3 Divide the same rows using Field

Science contains row 0: one E, so its entropy is 0. Arts contains rows 1 and 8: two U, so its entropy is also 0.

Step 4 Calculate entropy after the split

Each child gets a weight equal to its fraction of the parent rows:

$$H_{\text{after}} = \sum_v \frac{|D_v|}{|D|} H(D_v) = \frac{1}{3}(0) + \frac{2}{3}(0) = 0.$$

Step 5 Calculate information gain

Information gain is parent entropy minus weighted child entropy.

$$IG(D, A) = H(D) - \sum_v \frac{|D_v|}{|D|} H(D_v).$$

For this Field split, $IG = 0.9183 - 0 = 0.9183$ bits. The question removes all the uncertainty in these three training rows. **Check:** Degree is UG for all three rows. What would happen if we tried to split this subset using Degree?

★ KEY INSIGHT

Use the same parent rows when comparing features. When moving to a child, restart the calculation using only the rows in that child.

4 Compare the complete dataset using Degree

Return to all ten rows: 6 E and 4 U. Keep unrounded values while calculating; round displayed scores to four decimal places.

Step 1 Calculate the parent entropy

$$H(D) = -\frac{6}{10}\log_2\frac{6}{10} - \frac{4}{10}\log_2\frac{4}{10} \approx 0.9710.$$

Step 2 Divide the rows by Degree

ID3 makes one branch for each category. First record only the rows and class counts.

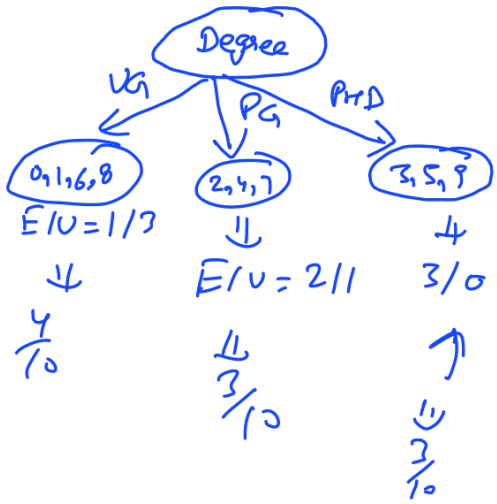
Degree	Rows	E / U
UG	0, 1, 6, 8	1 / 3
PG	2, 4, 7	2 / 1
PhD	3, 5, 9	3 / 0

Step 3 Calculate each child entropy

$$H(UG) = -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4} = 0.8113,$$

$$H(PG) = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.9183.$$

PhD has only E, so $H(PhD) = 0$.



Step 4 Calculate weight times entropy and complete the table

Each weight is the child's row count divided by the ten parent rows:

$$UG: \quad (4/10)H(UG) \approx 0.3245,$$
$$PG: \quad (3/10)H(PG) \approx 0.2755,$$
$$PhD: \quad (3/10)H(PhD) = 0.$$

Now collect all the calculated values in the full table.

Degree	Rows	E / U	Entropy	Weight times entropy
UG	0, 1, 6, 8	1 / 3	0.8113	0.3245
PG	2, 4, 7	2 / 1	0.9183	0.2755
PhD	3, 5, 9	3 / 0	0	0

Step 5 Combine the children and calculate gain

$$H_{\text{after}}(Degree) = \frac{4}{10}H(UG) + \frac{3}{10}H(PG) + \frac{3}{10}H(PhD) = 0.6000.$$

$$IG(D, Degree) \approx 0.9710 - 0.6000 = 0.3710 \text{ bits.}$$

Classroom check: why is the UG weight 4/10 rather than 1/3?

4 Continued compare Field and choose the root

Start again with **all ten rows**. Field has two categories: Science and Arts.

Field	Rows	E / U	Entropy	Weight times entropy
Science	0, 2, 3, 6, 9	4 / 1	0.7219	$(5/10)0.7219 = 0.3610$
Arts	1, 4, 5, 7, 8	2 / 3	0.9710	$(5/10)0.9710 = 0.4855$

Science:

$$H(Science) = -\frac{4}{5}\log_2\frac{4}{5} - \frac{1}{5}\log_2\frac{1}{5} = 0.7219.$$

Arts:

$$H(Arts) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.9710.$$

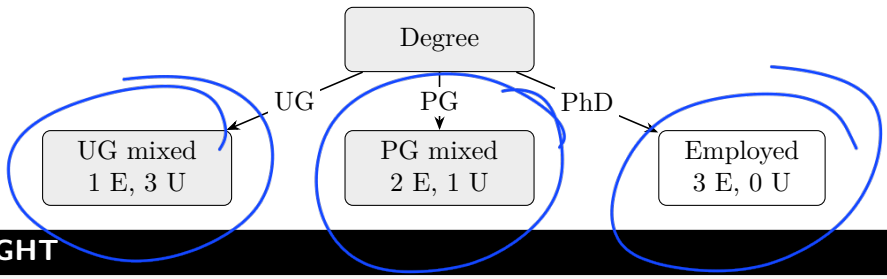
Using unrounded child values,

$$H_{\text{after}}(Field) = \frac{5}{10}H(Science) + \frac{5}{10}H(Arts) \approx 0.8464,$$

$$IG(D, Field) = H(D) - H_{\text{after}}(Field) \approx 0.1245.$$

Feature	Weighted child entropy	Information gain
Degree	0.6000	0.3710
Field	0.8464	0.1245

Choose Degree as the root: it has the larger information gain.



★ KEY INSIGHT

Both candidates used the same parent entropy. The larger gain therefore gives the smaller weighted child entropy. Continue the search only inside the mixed UG and PG groups.

5 Finish the ID3 tree and read its predictions

Split UG using Field

UG contains rows 0, 1, 6 and 8. Science has rows 0 E and 6 U; Arts has rows 1 U and 8 U.

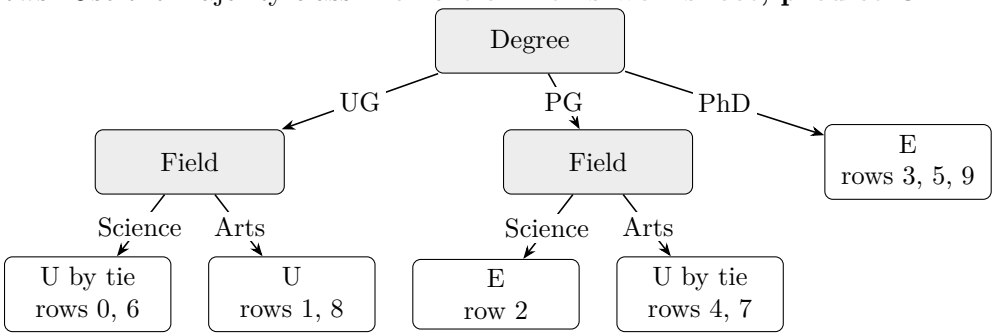
$$H_{\text{after}}(Field \mid UG) = \frac{2}{4}(1) + \frac{2}{4}(0) = 0.5, \quad IG \approx 0.8113 - 0.5 = 0.3113.$$

Split PG using Field

PG contains rows 2, 4 and 7. Science has row 2 E; Arts has rows 4 U and 7 E.

$$H_{\text{after}}(Field \mid PG) = \frac{1}{3}(0) + \frac{2}{3}(1) = 0.6667, \quad IG \approx 0.2516.$$

PhD is already pure and predicts E. In the mixed final groups, Degree and Field cannot separate the remaining rows. Use the majority class. **For a tie in this worksheet, predict U.**



Worked prediction: Degree = PG and Field = Science follows PG → Science → E. Degree = PhD reaches E without another question. **Check the training rows:** rows 0 and 7 are predicted U but actually have outcome E. The other eight rows are correct, so training accuracy is $8/10 = 80\%$. There are five leaves and the maximum depth is two.

★ KEY INSIGHT

Rows 0 and 6 have identical Degree and Field values but different outcomes. So do rows 4 and 7. No deterministic tree using only these two features can distinguish the rows within either pair.

Trace a new case: Degree = UG and Field = Science. Give its path and explain the prediction.

Practice P1 Entropy gain and weighting

P1a Match the calculation to its meaning

Match 1–4 to A–D. Each option is used once. Explain one match using the Degree split.

Calculation	Meaning
1. Parent entropy	A. Uncertainty left after all rows have been divided
2. One child's entropy	B. Uncertainty removed by a feature
3. Weighted child entropy	C. Class uncertainty before a question
4. Information gain	D. Class uncertainty within one group

P1b True or false justify and correct

1. Swapping the class names E and U changes the entropy.

2. If every child is pure, information gain equals parent entropy.

3. ID3 can separate rows 0 and 6 by splitting Degree or Field again.

P1c Dry run find a weighting mistake

A ten-row parent has 6 E and 4 U. A proposed split gives one child with 2 E and another child with 4 E and 4 U. Use parent entropy 0.9710.

(i) Find both child entropies, then their weighted average.

(ii) Find information gain.

(iii) A student uses $(0 + 1)/2 = 0.5$. Explain why this makes the gain too large.

Practice P2 Complete an ID3 dry run

P2a Compare the two categorical features

This is a separate six-row practice table using the same types of features and outcome labels. Use Degree and Field only. For tied leaf counts, predict U. Given entropies: $H(1/2) = 1$, $H(2/3) = 0.9183$, $H(3/4) = 0.8113$.

Row	Degree	Field	Job Outcome
A	UG	Science	U
B	UG	Arts	U
C	PG	Science	U
D	PG	Arts	E
E	PG	Arts	E
F	PG	Science	E

Count the parent E/U labels and calculate parent entropy. Fill the table. List Degree children as UG, PG and Field children as Science, Arts.

Feature	Child E / U counts	Weighted entropy	Gain
Degree	_____	____	____
Field	_____	____	____

Choose the root and show one weighted calculation.

P2b Recurse draw and predict

Within PG, split using Field. Calculate the local weighted entropy and gain. Draw the tree, then predict (PG, Science) and (UG, Arts).

P2c Explain the remaining error. Can these two features distinguish C from F? Calculate the final training accuracy.

PART III Compare CART and basic ID3

Both build trees from examples, but their split choices and supported prediction tasks differ.

6 CART and basic ID3 side by side

Property	CART	Basic ID3
Prediction task	Classification and regression	Classification only
Classification split rule used here	Choose the lowest weighted child Gini	Choose the largest information gain from entropy
Regression score	MSE for the squared-error example	Not a basic ID3 task
Branch structure	Always binary	One branch per categorical value
Numerical features	Can test thresholds	Does not directly use numerical features
Categorical example	Degree = UG: UG versus not UG	Degree: UG, PG and PhD
Leaf prediction	Majority class; mean for squared-error regression	Majority class, with a rule for ties
Pruning	Can use growth limits and cost-complexity pruning	No pruning step in the basic ID3 procedure shown here

Read the whole procedure

Using entropy as a scoring option does not automatically make a program ID3. For example, scikit-learn's classification tree still builds binary CART-style splits when its criterion is entropy. **Classroom check:** in our ID3 example, why does choosing Degree create three children, while the CART question Degree = UG created two?

♦ TAKEAWAY

Classification or regression describes the target. Categorical or numerical describes a feature. Basic ID3 needs a classification target and categorical features.

Optional independent study

C4.5 is a later extension of ID3 that can handle categorical and numerical features. It is **not covered in this lecture**. Interested students are encouraged to study it independently.

PART IV Overfitting and underfitting

A tree should learn patterns that help with new rows. A good training score alone does not tell us whether it has done that.

7 How much tree growth is useful

▷ HOOK

If we stop early, the tree may miss a useful question. If we keep adding questions, it may learn small details that appear only in the training data. We need to check both training and validation results.

Compare three versions of the employment tree

Tree	Predictions	Training accuracy
One leaf	Always predict E, the overall majority	$6/10 = 60\%$
Degree only	UG predicts U; PG and PhD predict E	$8/10 = 80\%$
Finished ID3 tree	Add Field inside UG and PG; use U for ties	$8/10 = 80\%$

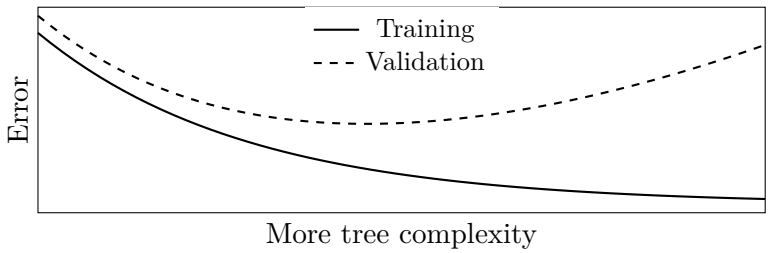
The Field questions reduce entropy, but the number of correct predictions stays the same. In the Degree-only tree, rows 0 and 4 are wrong. In the finished tree, rows 0 and 7 are wrong. Extra questions do not always increase training accuracy.

Underfitting

A depth limit can stop a tree before it asks a useful question. In our employment example, stopping at one leaf ignores the useful differences between Degree groups.

Overfitting

A tree can keep adding questions to isolate small training exceptions. It may create long paths and leaves containing just one or two rows, fitting an unusual outcome rather than a rule that works for new people. Training error may fall while validation error rises. More growth is therefore useful only when validation results support it.



These curves illustrate a possible pattern; they are not measured results from the employment dataset.

♦ TAKEAWAY

Use validation data to choose tree size. A deep tree is not automatically bad, and a small tree is not automatically good. Also, conflicting labels can prevent a perfect training fit even when we allow more growth.

Practice P3 Compare the methods and diagnose fit

P3a Match the tree to its method

Options: A = basic ID3, B = CART classification, C = CART regression. Use each option once and give one reason.

Tree-building rule	Match
1. Predict Test Score, use binary splits and compare weighted child MSE.	____
2. Predict Job Outcome, use one branch per Degree category and compare information gain.	____
3. Predict a class with scikit-learn, use the entropy criterion and still make binary splits.	____

P3b Choose a useful depth

A separate classification dataset gives these results. All three models use the same training and validation rows. Lower error is better.

Maximum depth	Training error	Validation error
1	40%	42%
3	15%	17%
8	0%	29%

Which depth would you select? Which results suggest underfitting and overfitting? Use the numbers to explain.

P3c True or false justify your answer

1. Basic ID3 directly accepts a numerical feature such as Average Grade.

2. Every additional question must improve training accuracy.

3. Pruning always improves validation accuracy.

PART V Pruning techniques

Growth limits act while the tree is being built. Post-pruning removes parts of a tree after growth.

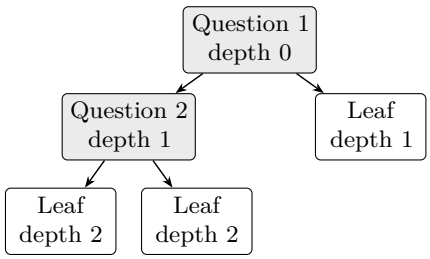
Pre-Pruning

Pre-pruning sets conditions that a proposed split must pass before it is added. The five settings below directly restrict tree growth. Feature subsampling is explained separately.
In each pair, the left sketch shows growth without the named limit. The right sketch shows the tree grown with that limit. These are small illustrative trees.

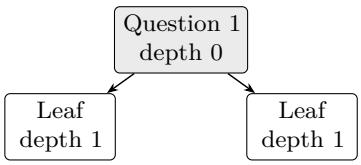
1 Maximum depth (`max_depth`)

This limits the number of edges from the root to a leaf. The root is at depth 0. With `max_depth = 1`, the root may split, but its children cannot ask another question.

Before applying the depth limit



After applying `max_depth = 1`

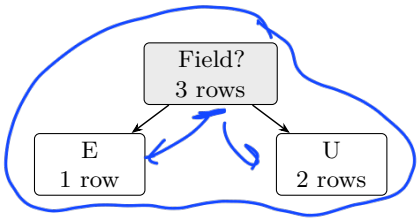


The second question is never added in the limited tree. Both remaining leaves are at depth 1.

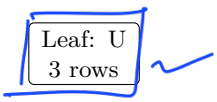
2 Minimum split size (`min_samples_split`)

This sets the minimum rows a **parent node** needs before a split can be attempted. With a limit of 4, the three-row node below cannot split, even though its question would separate the classes.

Before applying the parent-size limit



After applying `min_samples_split = 4`



Parent is too small:
$3 < 4$

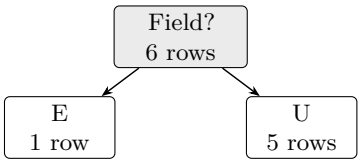
The unsplit node predicts U, the majority of its three training labels. A sufficiently large parent must still pass the other growth rules.

Pre-Pruning continued

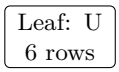
3 Minimum leaf size (`min_samples_leaf`)

This sets the minimum rows allowed in **each child** created by a split. With a limit of 2, the proposed 1/5 split below is rejected. If no other candidate passes the rules, the six-row node stays a leaf, as shown.

Before applying the child-size limit



After applying `min_samples_leaf = 2`



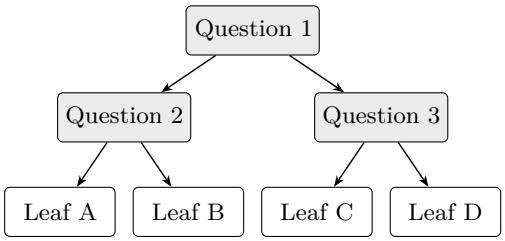
The one-row child would fail the minimum leaf size of 2

A different proposed split into 3 and 3 would pass this size check. Minimum split size checks the parent; minimum leaf size checks the children.

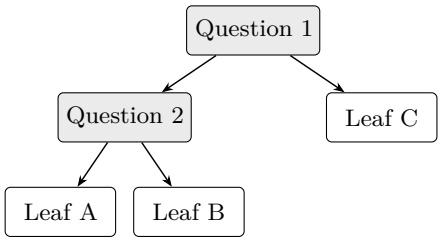
4 Maximum leaves (`max_leaf_nodes`)

This limits the total number of leaves. Suppose the left split is chosen first. With a three-leaf budget, Question 2 can be added, but Question 3 cannot: it would create a fourth leaf.

Before applying the leaf budget
Four leaves



After applying `max_leaf_nodes = 3`
Three leaves



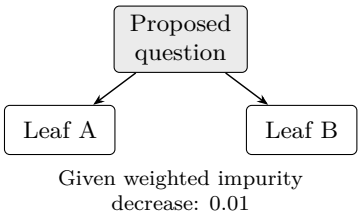
In a binary tree, splitting one leaf into two increases the total leaf count by one. The leaf budget limits the whole tree, rather than the length of one path.

Pre-Pruning continued

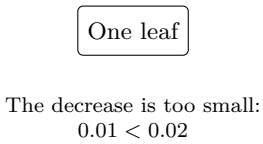
5 Minimum impurity decrease (`min_impurity_decrease`)

A split must reduce impurity by at least the specified amount. The software weights this improvement by the share of training rows reaching the node.
Suppose the proposed question gives a weighted decrease of 0.01, but the setting requires 0.02. That question is not added. If no other candidate meets the rules, the node remains a leaf.

Before applying the improvement threshold



After applying `min_impurity_decrease = 0.02`



Advantages of pre-pruning

It saves time because the tree does not grow branches that will later be removed. It often creates a smaller, easier-to-read tree and can reduce the chance of fitting tiny training details.

Disadvantages of pre-pruning

It can stop too soon and miss a useful later question. Very strict limits can cause underfitting. Choose the limits using validation results rather than assuming one setting works for every dataset.

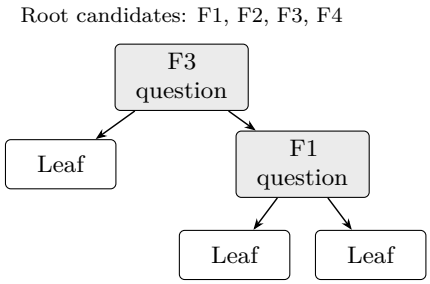
♦ TAKEAWAY

Depth, parent size, child size, total leaves and impurity improvement check different conditions. A proposed split must pass every relevant condition.

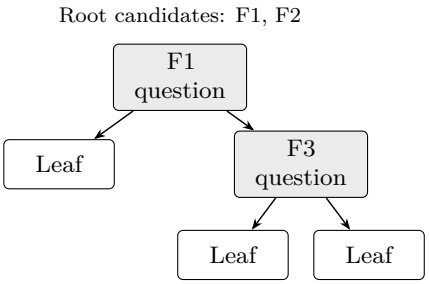
Feature Subsampling (`max_features`)

Limits how many predictors are considered while searching for a split. It changes the available split candidates and can act as a form of regularization, but it does not directly bound the depth or number of leaves and is not, strictly speaking, a pruning criterion.
Imagine four predictors, F1–F4. At the root, considering only F1 and F2 can change the chosen question. At a later node, a different subset, such as F3 and F4, may be considered.

Before feature subsampling
All four root candidates



After feature subsampling
Two root candidates



Both possible trees above have depth 2 and three leaves. A tree using feature subsampling can still grow deep and have many leaves.

Practice P4 Growth limits and feature subsampling

(a) Match. A = maximum depth; B = minimum split size; C = minimum leaf size; D = maximum leaves; E = feature subsampling; F = minimum impurity decrease. Use each option once.

Situation during growth or split search	Match
1. The parent contains too few rows to try splitting.	____
2. One child would contain too few rows.	____
3. Another question would make the path too long.	____
4. Another split would exceed the total leaf limit.	____
5. The search considers a smaller set of predictors.	____
6. The weighted impurity improvement is too small.	____

(b) Dry run. Treat the trials independently; all other checks pass.

(i) Parent size 6, children 1 and 5, minimum split size 4 and minimum leaf size 2. Allow or reject?

(ii) Same settings, but children 3 and 3. Allow or reject?

(iii) A leaf is at depth 1 and maximum depth is 1. Can it split?

(iv) A binary tree has 4 leaves and the maximum is 5. Can one leaf split into two?

PART VI Post-Pruning

Compare an already grown tree with simpler versions.

Post-Pruning

First grow the tree, then simplify it by replacing selected subtrees with leaves. Check which simpler version is worth keeping. We will use cost-complexity pruning and reduced-error pruning.

9 Cost-complexity pruning

▷ HOOK

A large tree may fit the training data well, but it has many leaves. A smaller tree may fit slightly less well and be easier to use. How can we compare these two things together?

Add a penalty for every leaf:

$$\text{Cost}(T) = R(T) + \alpha L(T).$$

- $R(T)$ is the training score. Smaller means a better fit to the training data.
- $L(T)$ is the number of leaves.
- α is the penalty charged for each leaf. It is zero or positive.

For one chosen α, prefer the tree with the **smallest total cost**. In scikit-learn, the training score is the sum of weighted leaf impurities; it is not validation accuracy or a count of correct predictions.

Worked example compare four tree sizes

The scores below are given for a separate teaching example. These trees are not fitted to the ten employment records.

Tree	Leaves	Training score	Cost at .005	Cost at .010	Cost at .020
T_8	8	.10	.140	.180	.260
T_5	5	.12	.145	.170	.220
T_2	2	.15	.160	.170	.190
T_1	1	.20	.205	.210	.220

At $\alpha = .005$: T_8 costs $.10 + .005(8) = .140$. This is the smallest cost, so choose T_8. **At $\alpha = .010$:** T_5 and T_2 both cost .170. Our tie rule chooses fewer leaves, so choose T_2. **At $\alpha = .020$:** T_2 costs $.15 + .020(2) = .190$, which is the smallest cost.

What does a larger alpha do

A larger alpha makes each extra leaf more expensive. It therefore favors a smaller tree. At .060, the four costs are .580, .420, .270 and .260; the single root leaf has the smallest cost.

♦ TAKEAWAY

Cost-complexity pruning balances training fit and tree size. A very large penalty can remove too much and cause underfitting. Use validation results to choose alpha.

10 Which branch should we remove first

▷ HOOK

Removing a branch may increase the training score. We ask: how much does the score increase for each leaf we save? A smaller increase means the branch is cheaper to remove.

Calculate the cost of removing a branch

Replacing a branch with one leaf gives

$$\text{Extra training score} = \text{score as one leaf} - \text{current branch score},$$

$$\text{Leaves removed} = \text{current leaves} - 1.$$

Divide the two quantities:

$$\alpha_{\text{eff}} = \frac{\text{extra training score}}{\text{leaves removed}}.$$

This value is called the branch's **effective alpha**. Remove the branch with the smallest value first. This is called **weakest-link pruning**.

Branch	Score as one leaf	Current score	Current leaves	Effective alpha
A	.08	.02	4	$(.08 - .02)/3 = .020$
B	.06	.05	2	$(.06 - .05)/1 = .010$

Worked decision: remove B first. Its score increases by .01 while one leaf is saved. A's score increases by .06 while three leaves are saved: .02 for each leaf. After removing a branch, calculate the remaining scores and leaf counts again.

How do we choose alpha for the final tree

Try candidate alpha values and compare their **validation accuracy**. For example, suppose the following validation results are obtained:

Alpha	Leaves	Validation accuracy
.005	8	84%
.010	2	88%
.060	1	60%

Choose .010 from these results because 88% is the highest validation accuracy. These are illustrative results, not measurements from the employment dataset.

♦ TAKEAWAY

Training rows build the tree. Validation rows help choose its size. Test rows check the final chosen model once. Do not choose alpha by repeatedly checking the test set.

Practice P5 Cost-complexity pruning dry run

P5a Calculate costs at three new alpha values

Use the same example trees: T_8 has 8 leaves and training score .10; T_5 has 5 and .12; T_2 has 2 and .15; T_1 has 1 and .20. For each alpha column, calculate training score + alpha × leaves. Choose the smallest cost in that column.

Tree	$\alpha = .008$	$\alpha = .012$	$\alpha = .055$
T_8	____	____	____
T_5	____	____	____
T_2	____	____	____
T_1	____	____	____
Choose	____	____	____

Show one calculation. At what alpha do T_5 and T_2 have equal cost? Use fewer leaves to break a tie.

P5b Decide which branch to remove

Use the given scores to calculate effective alpha.

Branch	**Score as one leaf**	**Current score**	**Current leaves**	**Effective alpha**
C	.14	.08	4	____
D	.09	.08	2	____

Which branch would you remove first? Explain why the larger branch is not automatically removed first.

P5c True or false explain your choice

1. The model with the best training score must also have the best validation score.

2. A very large alpha can leave only the root leaf.

11 Reduced-error pruning

▷ HOOK

A tree gets 17 of 20 validation rows correct. After removing a branch, it still gets 17 correct. We can keep the simpler tree because its validation accuracy has not fallen.

Reduced-error pruning checks whether removing a branch makes validation performance worse. We replace the branch with one leaf and compare predictions before and after the change.

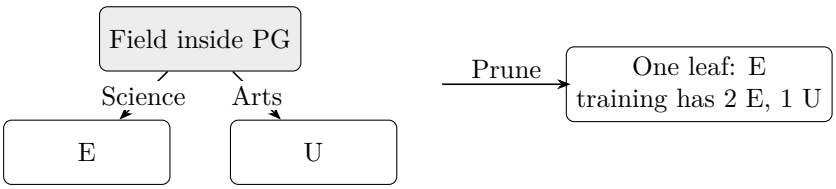
Follow these steps

1. Start with a grown tree and a separate validation set. Check the deeper branches before their parents.
2. Choose a branch to test. Find the majority outcome among the **training rows** at that branch. Use U if the counts tie.
3. Temporarily replace the branch with one leaf predicting that outcome.
4. Predict all the validation rows again and calculate accuracy.
5. If accuracy is the same or higher, keep the change. If accuracy falls, undo it. Compare the next trial with the tree you have now kept.

Worked example on the same 20 validation rows

Tree being checked	Correct	Accuracy	Decision
Current tree	17	85%	Starting score
Remove branch A	17	85%	Keep the change
Then try removing B	16	80%	Undo this change

A is kept because 85% stays 85%. B is restored because 80% is worse than 85%. An improvement is welcome, but a tie is enough to keep a smaller tree.



The drawing shows a possible replacement in the PG group. The 20-row scores above are a separate teaching example.

Compare the pruning choices

Method	When it acts	How it decides
Pre-pruning	While the tree grows	Apply limits such as depth and leaf size
Cost-complexity pruning	After growth	Compare training score plus a leaf penalty; choose alpha with validation
Reduced-error pruning	After growth	Keep a removal if validation accuracy does not fall

♦ TAKEAWAY

Training labels choose the replacement leaf's class. Validation labels check the change. If using an error score rather than accuracy, keep a change only when the error stays the same or falls.

Practice P6 Reduced-error pruning dry run

P6a Replace the branch and count correct predictions

A node has 3 E and 1 U in training. Six validation rows reach this node. The current branch predicts as shown below. The other 14 validation rows have 12 correct predictions; they are unchanged by this trial.

Row	Actual	Current prediction	One-leaf prediction	Leaf correct
1	E	E	____	____
2	E	E	____	____
3	U	U	____	____
4	E	U	____	____
5	E	E	____	____
6	E	U	____	____

Fill the replacement predictions. Calculate accuracy before and after pruning using **all 20 rows**. Keep or undo the change? Which data chose the replacement class?

P6b Compare each trial with the current tree

Start from the accepted tree with 17 of 20 correct above. Each trial below is on the tree currently kept at that step. Accept a tie.

Trial	Correct after trial	Keep or undo	Correct in the tree kept
B	16	____	____
C	18	____	____
D	17	____	____
E	18	____	____

Why is D undone even though 17 correct was acceptable at the start?

P6c True or false: the replacement leaf should predict the majority class in the validation rows.

PART VII Closing practice

Use validation results to choose tree size. Keep the final test set for checking the chosen model.

Practice P7 Choose a model and correct a mistake

P7a Choose a classification tree

For a separate classification dataset, four candidate trees give these results. Choose the highest validation accuracy. If there is a tie, choose fewer leaves.

Model	Leaves	Training accuracy	Validation accuracy
M1	8	100%	84%
M2	5	98%	88%
M3	2	92%	88%
M4	1	60%	58%

Which model should be chosen? Use the table to explain why the largest training accuracy and the smallest tree are not enough on their own.

P7b Make the same decision for regression

Three regression trees have the following scores. Lower MSE is better.

Model	Training MSE	Validation MSE
G1	1	20
G2	5	8
G3	15	18

Choose a model and explain which score you used.

P7c Fix a wrong procedure

A student trains a tree using every row, tries many alpha values on the test set and reports the best test accuracy. Explain what went wrong and write the correct order of training, validation and testing.